# Maxwell Morrison

**B.S. Candidate in Electrical Engineering and Applied Mathematics**

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Undergraduate researcher at the University of Delaware with a background in electrical engineering and applied mathematics. Research interests include signal processing, computational imaging, machine learning for sensing systems, embedded systems, and computational neural information engineering.

Last updated July 2026

## EDUCATION

### University of Delaware

**Expected May 2027**

Newark, DE

- B.S. Electrical Engineering, College of Engineering
- B.S. Applied Mathematics, College of Arts and Sciences
- GPA: 3.6

## RESEARCH INTERESTS

Signal processing; computational imaging; kernel methods; submodular optimization; machine learning for sensing systems; information-theoretic objectives; embedded perception; computational neural information engineering; applied and computational mathematics.

## RESEARCH EXPERIENCE

### Undergraduate Researcher
**Summer 2026**

Computational Neural Information Engineering Laboratory (CNIEL), University of Delaware | Newark, DE

- Conducting undergraduate research under Dr. Austin Brockmeier in the Computational Neural Information Engineering Laboratory.
- Summer project: Maximizing von Neumann Entropy for Submodular Semi-Supervised Data Augmentation.
- Encoding images as DINO feature representations and selecting unlabeled examples to augment labeled training data through a von Neumann entropy objective with a scaled log-probability soft penalty.
- Maintaining research code repository: github.com/maxbmorr/DINO-Kernel-Herding

### Undergraduate Researcher
**Jan. 2025 - Aug. 2025**

Underwater Acoustics Research, University of Delaware | Newark, DE

- Worked with Dr. Mohsen Badiey and a team of undergraduate researchers on an electronic acoustic recorder.
- Developed an embedded sensing system using the ESP-WROOM-32D with custom hardware and firmware.
- Contributed to embedded system development in support of underwater acoustics research.

## PRESENTATIONS

### Maximizing von Neumann Entropy for Submodular Semi-Supervised Data Augmentation
**Summer 2026**

Computational Neural Information Engineering Laboratory, University of Delaware

- Presented summer research poster with Zhi Li and Dr. Austin J. Brockmeier.
- Developed a semi-supervised data augmentation approach using DINO image embeddings, von Neumann entropy, and a scaled log-probability soft penalty.

### DroneEye: LiDAR- and Machine-Learning-Based Drone Detection System
**2026**

University of Delaware Research Day

- Presented poster and technical presentation on a drone detection system using computer vision, machine learning, and LiDAR-based depth perception.
- Received the Research Day People's Choice Award for DroneEye.

## AWARDS

- 2026 Research Day People's Choice Award Winner for DroneEye, Department of Electrical and Computer Engineering, University of Delaware
- Institute of Electrical and Electronics Engineers Delaware Bay Section Student Activities Award, Department of Electrical and Computer Engineering, University of Delaware
- Eagle Scout, 2021
- Dean's List

## TEACHING AND ENGINEERING EXPERIENCE

### Supplemental Instructor / Teaching Assistant
**Jan. 2025 - Present**

University of Delaware | Newark, DE

- Supplemental Instructor for CPEG202, Introduction to Digital Logic, under Dr. Nathan Lazarus and Mr. Thomas Lum.
- Hold weekly student support hours and assist students with course concepts, lab work, and problem solving.
- Support instruction through grading quizzes, exams, and course materials while serving as a primary contact for students.

### Lab Assistant
**Jan. 2025 - Present**

University of Delaware | Newark, DE

- Lab Assistant for ELEG205, Introduction to Analog Circuits, under Dr. Fouad Kiamilev.
- Lead weekly lab sessions and support students in circuit analysis, measurement techniques, and lab procedures.
- Assist with grading lab coursework and help ensure smooth operation of instructional lab activities.

### Electrical Engineering Intern
**May 2025 - Aug. 2025**

Henry Adams | Baltimore, MD

- Worked in Revit and AutoCAD to develop detailed electrical drawings for contractors.
- Performed lighting simulations using AGi-32.
- Contributed to arc flash studies using SKM.

## SELECTED PROJECTS

### DroneEye

Senior Design, Computer Vision, LiDAR

Computer-vision and machine-learning drone detection system using LiDAR for depth perception and a custom-trained model to detect drones and estimate Cartesian coordinates through a graphical interface. Presented at University of Delaware Research Day.

### Diabeasy

UD Hackathon, Python

Encrypted Tkinter application for tracking insulin usage and calculating dosage amounts.

### PitchMAX

Hardware, Firmware

Lower-cost pitch calling system inspired by PitchCom, using custom hardware and firmware to communicate called pitches between players.

### PyroPlan

UD Hackathon, Web App, Simulation

Web platform for modeling fire spread in level 1 materials server-side, making simulation accessible without local software installation. Repository: github.com/maxbmorr/2026-Hackathon.

### Mom Blood Sugar Tracker

Healthcare, Web App

Web platform that lets multiple caregivers submit orders and coordinate blood sugar and insulin-related care.

## SELECTED COURSEWORK

### Electrical Engineering

- ELEG479, Introduction to Biomedical Imaging - A

### Computing / Computational Methods

- ELEG467, Introduction to Computational Methods - A
- MATH426, Computational Mathematics - In Progress

### Applied Mathematics

- MATH503, Complex Analysis - In Progress
- MATH508, Advanced Calculus with Nonlinear Dynamics - In Progress

## TECHNICAL SKILLS

- Signal processing and computational methods
- Machine learning and computer vision
- Embedded systems and firmware
- Digital logic and analog circuits
- Python
- Tkinter
- MATLAB
- Git / GitHub
- Revit
- AutoCAD
- AGi-32
- SKM
- ESP-WROOM-32D

## AFFILIATIONS

- IEEE Student Chapter
- IEEE Eta Kappa Nu
- Troop 97 (Scouts BSA)